Name ______________________________ Date ______________

10.1 Logic expressions

Getting started

Complete the truth tables for the following gates:

1 AND

X	Y	X AND Y
0	0	
0	1	
1	0	
1	1	

2 OR

X	Y	X OR Y
0	0	
0	1	
1	0	
1	1	

3 NOT

X	NOT X
0	
1	

4 NAND

X	Y	X NAND Y
0	0	
0	1	
1	0	
1	1	

5 NOR

X	Y	X NOR Y
0	0	
0	1	
1	0	
1	1	

6 XOR

X	Y	X XOR Y
0	0	
0	1	
1	0	
1	1	

Practice

1 Evaluate each statement to identify if the output is 1 (true) or 0 (false).

a 1 AND 0

b 1 OR 1

c 1 NOR 0

d 1 XOR 1

e NOT(1)

f 1 NAND 0

g 1 AND 1

h 0 OR 0

i 0 NOR 0

j 1 XOR 0

2 Complete the truth table for the following logic expressions.

a

A	B	C	A AND B AND C
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

b

A	B	C	A XOR (B OR C)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

c

A	B	C	(A AND B) OR (C XOR B)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

d

A	B	C	(NOT A NAND B) NOR C
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

e

A	B	C	(C XOR A) AND (B OR A)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Challenge

Write a logic expression that makes use of **four** inputs; A, B, C and D.

Create the truth table for your logic expression.

Swap the expression with another learner and draw the truth table for their logic expression.

Compare your truth tables and see if they match.

Name ______________________ Date ____________

10.2 Logic circuits

Getting started

Draw each of the logic gate symbols.

1 AND

2 OR

3 NOT

4 NAND

5 NOR

6 XOR

Practice

1 Draw the logic circuit for the following logic statements:

  a A AND B

  b A OR NOT B

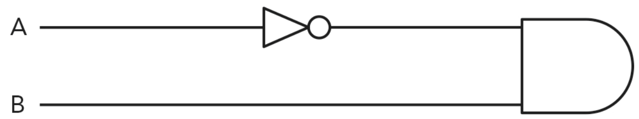
  c A AND B AND C

  d (A XOR B) OR C

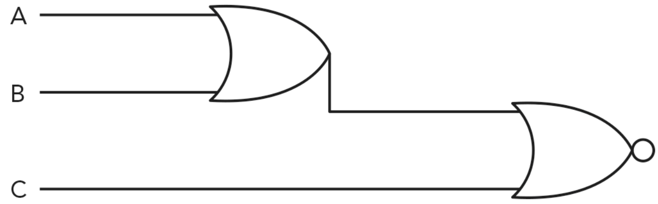
  e A NAND B NAND C

2 Write the logic expression for the following logic circuits

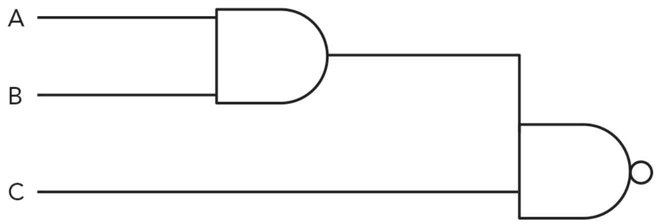
  a



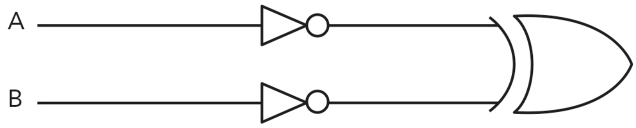
  b



c



d



Challenge

Write the logic statement for each of the truth tables **and** draw its logic circuit.

1

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

2

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

3

A	B	C	Output
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

4

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Name ________________________________ Date ______________

10.3 Logic

Getting started

Complete the following sentences:

The ____________________ gate only returns 1 when both inputs are 1.

The ____________________ gate only returns 0 when both inputs are 0.

The ____________________ gate only returns 1 if the input is 0.

The ____________________ gate returns 0 if both inputs are 0, and when both inputs are 1.

The ____________________ gate only returns 1 if both inputs are 0.

Practice

Correct the error(s) in the truth tables.

1 A AND (B OR C)

A	B	C	A AND (B OR C)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

2 A XOR NOT B

A	B	C	A XOR NOT B
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

3 (A NOR B) OR (C AND A)

A	B	C	(A NOR B) OR (C AND A)
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Challenge

You only need to work with a logic diagram, or expression, that has one output. They can, however, have more than one output.

Draw a logic diagram for the following 2-output statements.

1 X = A AND B

Y = C OR D

2 P = NOT (A AND B)

Q = (A XOR B) OR C

3 S = (A NOR B) AND C

T = NOT(A OR B)